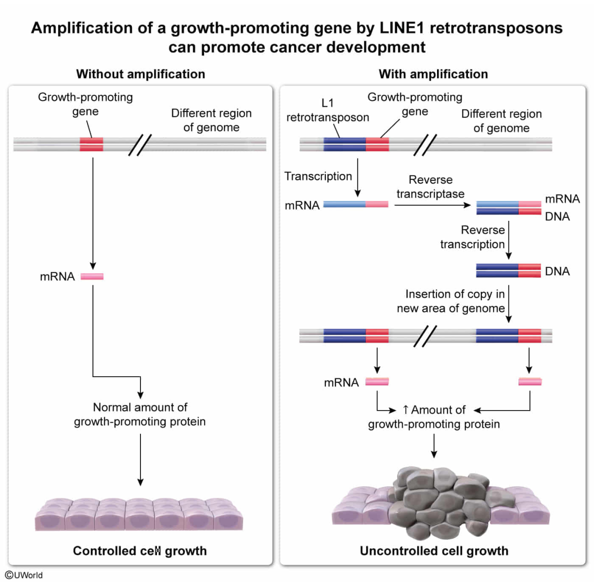
Seventeen percent of the human genome consists of repetitive sequences called LINE-1 (L1) retrotransposons, a small fraction of which can code for their reinsertion elsewhere in the genome. Such reinsertions can alter genome organization in various ways and are present in roughly half of human tumors. Given this, which of the following represents a means by which L1 retrotransposons are likely to promote cancer?

- A. Fusion of an inhibitory sequence to an oncogene [6%]
- B. Amplification of a growth-promoting gene [78%]
- C. Placement of a tumor suppressor gene under the control of a new, more active promoter [11%]
- D. Unequal crossing over leading to the deletion of multiple copies of a growth-promoting gene [3%]

Correct
79% answered correctly

Explanation:



DNA sequences that code for mRNA, rRNA, or tRNA make up a tiny fraction of the human genome. More than half of the genome consists of **repetitive sequences**, some of which are **transposable elements** that can be moved or copied to other locations (sometimes referred to as "cut-and-paste" and "copy-and-paste" mechanisms, respectively) in the genome. Cut or copied transposon sequences can be inserted into new locations on the same chromosome or on different chromosomes.

Most eukaryotic transposable elements are **retrotransposons**, which use a copy-and-paste mechanism involving an RNA intermediate to create additional copies of the retrotransposon DNA sequence in the genome. The RNA intermediate is reverse transcribed into DNA and inserted into the genome at the new location.

Retrotransposons can have varied effects on coding sequences or regulatory

noncoding sequences, depending on their size, number of copies, and insertion location(s). For example, inserted retrotransposon sequences may result in deletion or insertion of a chromosomal segment in a way that alters the expression or regulation of genes near the insertion. In addition, **nearby genes** are **sometimes copied along with** the **retrotransposon**, which **can amplify the gene** and/or **place** it **in** a **differently regulated environment** in the genome.

The question notes that L1 retrotransposons are common in the human genome and asks for a means by which such retrotransposons might promote cancer. When cells become cancerous, they lose one or more of the regulatory mechanisms that typically prevent the excessive growth characteristic of cancer cells. Therefore, a retrotransposon that causes **multiple copies of** a **growth-promoting gene** to be created is one **likely to promote cancer**.

(Choice A) Fusion of a stimulatory (*not* inhibitory) sequence to an oncogene could promote cancer. For example, fusion of a DNA sequence that alters the regulation of a transcription factor by phosphorylation or a specific hormone can alter the effect of the transcription factor on gene transcription.

(Choice C) Placement of a tumor suppressor gene under the control of a new, more active promoter would likely *suppress* rather than promote cancer development.

(Choice D) Deletion of a growth-promoting gene would likely *inhibit* the uncontrolled excessive growth that characterizes cancerous cells.

Educational objective:
Transposable elements are DNA sequences that can move or be copied to different locations in the genome. Transposable elements such as retrotransposons can result in multiple copies of genes being created as well as placing genes in new, differently regulated genomic regions.

Time spent:0 sec
Subject: Biology

QID:403894
System: Molecular Biology